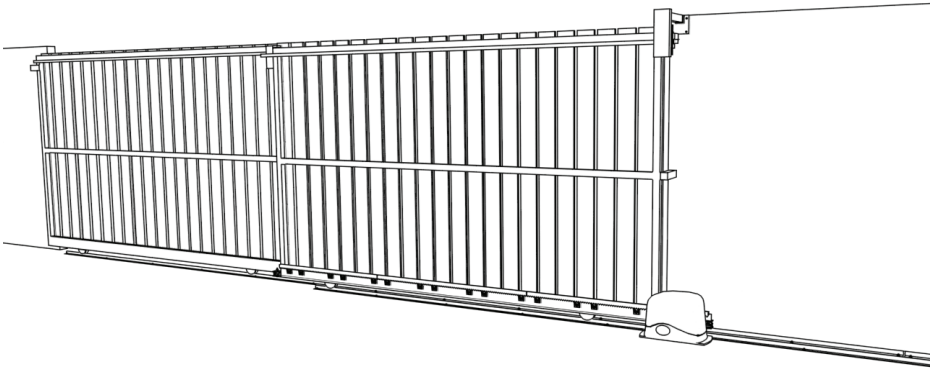
INSTALLATION MANUAL

FOR

Telescopic Sliding Gate with Aluminium Frame for Timber Cladding opening from left to right (viewed looking out towards the road)



Grant's Automation

Developers and Manufacturers
of Automation Technology
www.grantsautomation.com.au

IMPORTANT SAFETY INSTRUCTIONS!

CONSIDERING THE GENERAL PUBLIC:

When Installing an Automatic Gate that will be entered from a public road way, make sure the Gate is placed far enough from the road to prevent traffic congestion.

The Gate must be installed in a location that provides adequate clearance between it and adjacent structures when opening and closing to reduce risk of entrapment.

Install the Gate Operator on the inside of the property and fence line. **DO NOT** install an opener on the outside of the gate where the public has access to it.

The Gate and Gate Operator must comply with any applicable local council regulations.

CONSIDERING THE USERS:

If using the Auto-close feature it is highly recommended that a Point to Point Photo Electric Safety Beam (Photocell) is installed to prevent the gate closing on any vehicle using the gate.

It is also recommended that a seperate small side gate is used for pedestrians particularly if there will be children, disabled or elderly people using the gate.

If push buttons, key switches or Digital Keypads are installed, they should be within sight of the gate but not placed so the user will be tempted to reach through the gate to activate the gate operator.

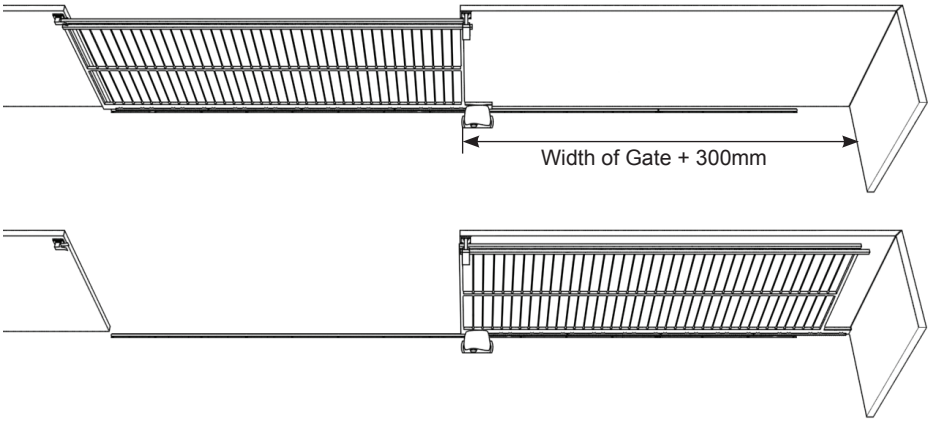
USER AWARENESS:

It is important to make sure everyone that will be using the gate is aware of the following dangers associated with automatic Gates: do not contact any part of the gate or walk in the path of the gate while it is moving. Never let children play with the gate controls. Do not attempt to “beat the gate” while it is closing. This is extremely dangerous.

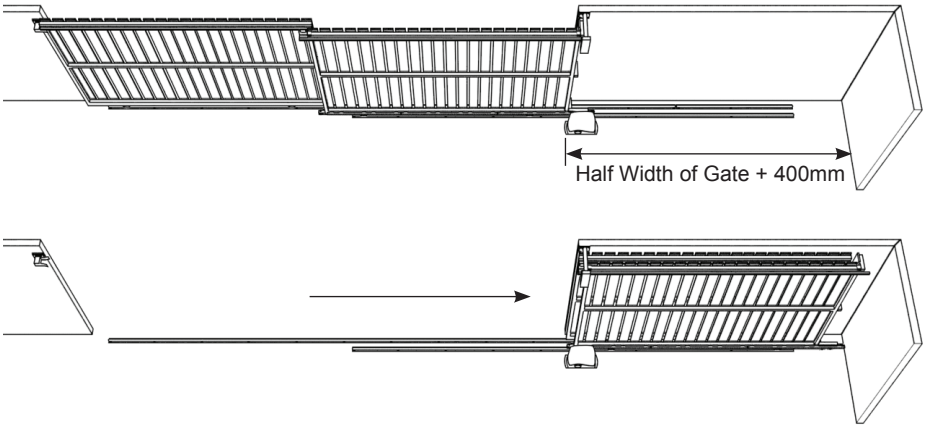
In the event you sell the property, make sure the new owners have a copy of these instructions. If you lose the instructions they can be downloaded from: www.grantsautomation.com.au.

HOW A TELESCOPIC GATE WORKS

Normally a sliding gate requires the whole width of the gate plus a bit more space to slide into.

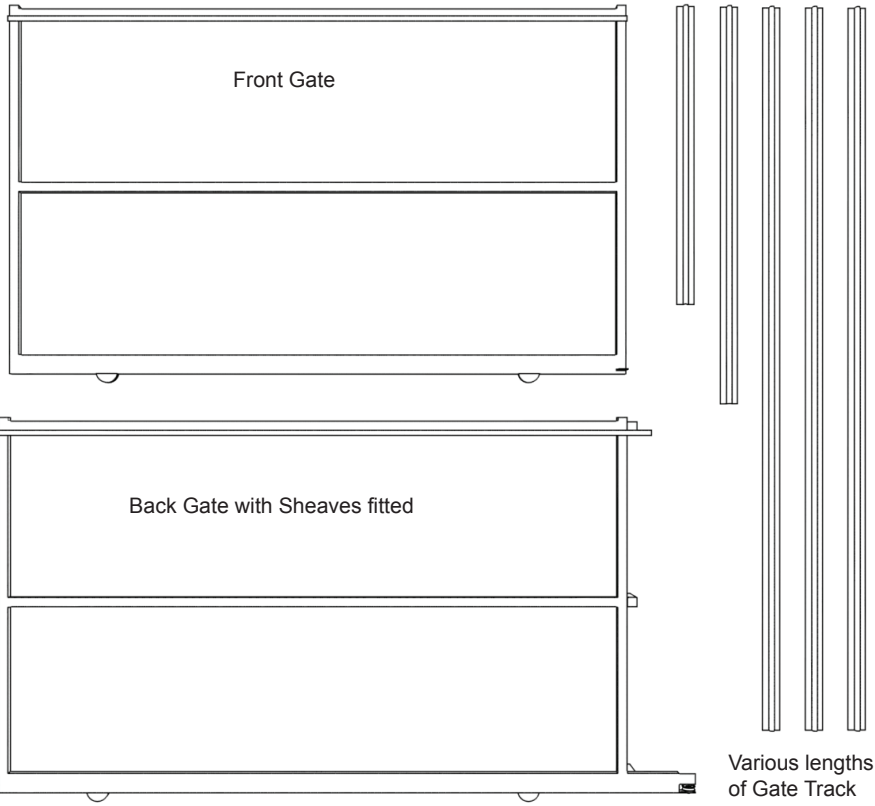


A telescopic Gate however is in two pieces with one piece sliding in front of the other so requires only half the gate width plus a bit more space to slide into allowing a 6m wide sliding gate to easily fit across a 10m wide front boundary.



Another advantage of a telescopic sliding gate is it requires only one automatic operator to move the back gate and the front gate is moved by wire rope from the back gate. The front gate travels at twice the speed of the back gate so a telescopic gate opens in half the time of a single piece gate. Installation of cabling is the same regardless.

TELESCOPIC GATE KIT INCLUDES




Keeper


Guide Mounting Bracket


Wire Rope Ground Connecting Bracket


2 x Nylon Guide Blocks

4x Gate Wheels with 8x metal tek screws.
2 x M10 x 50mm Dyna Bolts for wire rope ground connector.
Metal Pin Anchors for Gate track.
2 x lengths of Wire Rope.
4 x Grips for wire rope.
8 x Metal or Timber drilling Tek screws or Pin anchors for Keeper and Guide Bracket
2 x Metal drilling Tek screws for Guide stops
6 x Plastic end caps for gate frame
1 x Gate operator with key release and remote controls
Racking for automatic gate operator including metal drilling Tek screws to attach to the gate

TOOLS AND HARDWARE REQUIRED

THE TOOLS YOU'LL NEED INCLUDE:

A basic set of hand tools will be needed including: A hammer, spirit level, 4x "F" clamps, tape measure, marking pen and blocks of wood are also very handy.

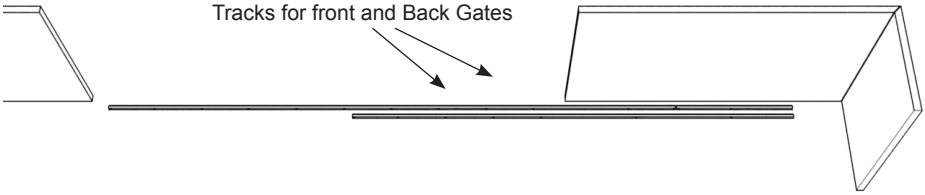
You'll also need an elecric drill with hammer action and variable speed control, a 6.5mm masonary drill bit, an 8mm metal drill bit, No.2 phillips screw driver tip and socket bit for 3/8" tek screws. An angle grinder is also handy although a hacksaw will do if you don't have one.

OTHER REQUIREMENTS BEFORE STARTING

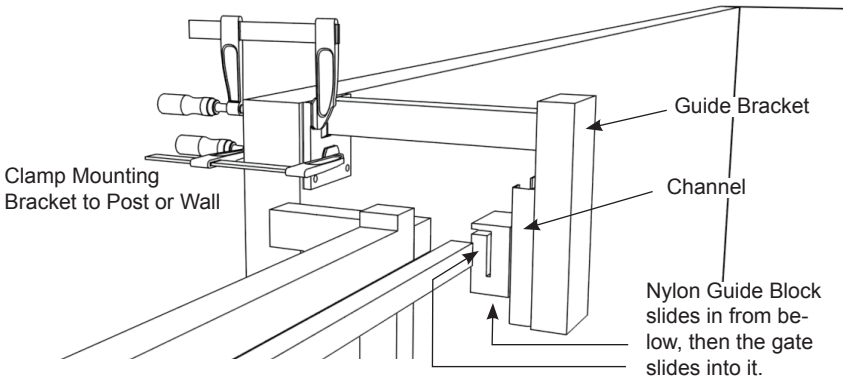
You will need suitable posts for the guide and keeper and a concrete footing for the track the gate will roll on. Details of these and a drawing will be provided with the kit.

INSTALLING THE GATE

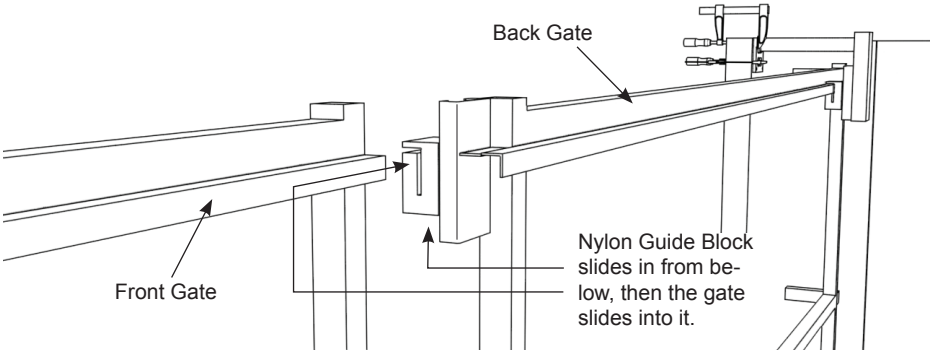
Lay the tracks down according to the supplied drawing but don't fix them in place yet



Then using 'F' clamps, clamp the guide bracket in place temporarily as per the the supplied drawings. Place the back gate on it's track then install the nylon guide block to guide the gate while sliding. ***Be careful not to allow the gate to slide back out of the nylon block otherwise it could fall over and become damaged or harm anyone in its way.***

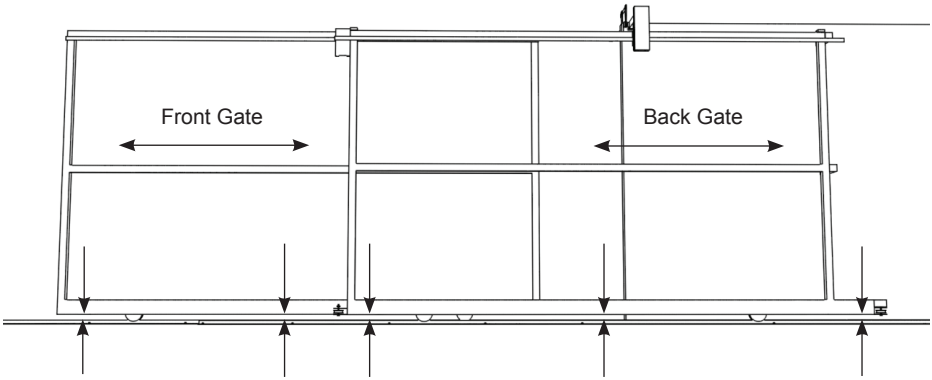


Now place the front gate on it's track and install the other nylon guide block between the gates allowing the front gate to slide into the back gate. ***Again be carefull not to allow it to slide back out again.***

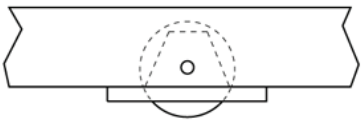


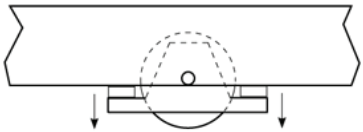
INSTALLING THE GATE

Check both gates run freely for the entire opening without either gate contacting their track, each other, posts or fencing.

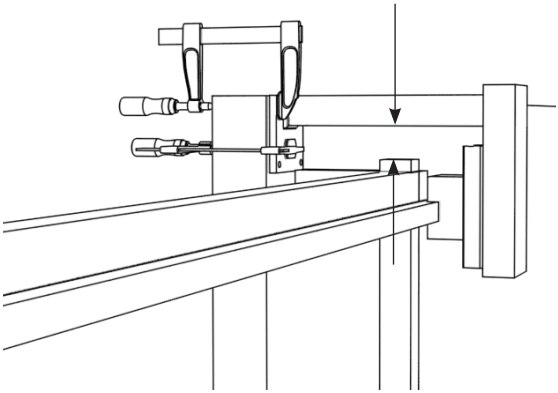


If either gate does touch the track and the track can't be set down any further once fixed to the concrete footing you can pack the wheels out with washers to raise the height of the gate enough to clear.



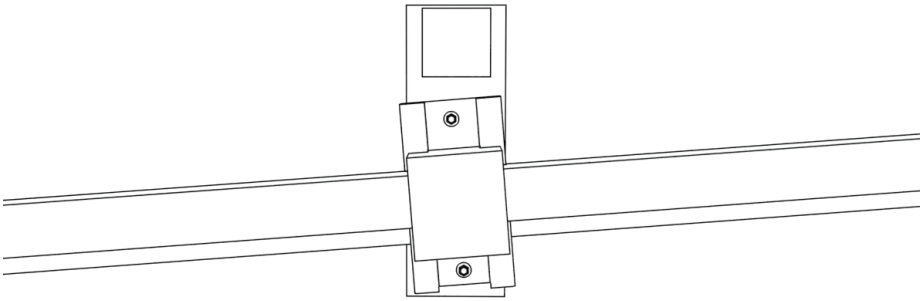


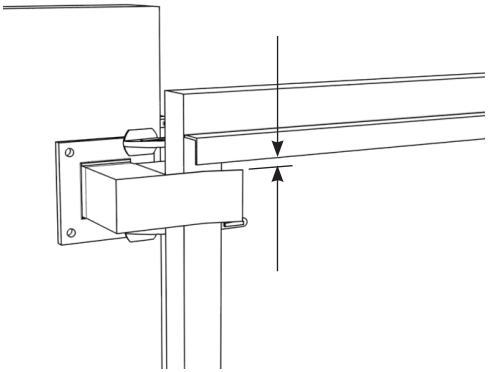
Check each gate has suitable clearance from the guide bracket allowing for any overlap of the timber cladding. You can move the guide bracket up a little to allow more clearance for timber but too much may cause the nylon guide block to come off the bottom of its channel.



INSTALLING THE GATE

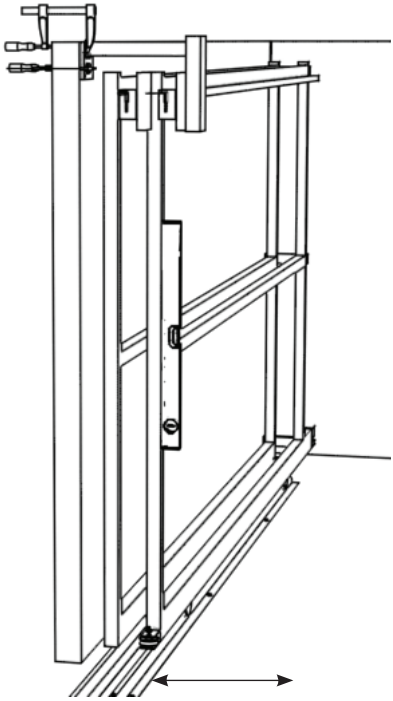
If the top rail of the gate is raked (sloped) then the guide blocks and channels will be on the same angle as the top rail to prevent binding.





Clamp the keeper in place as per the supplied drawing and make sure the guide rail on the front gate clears it. If it doesn't just move it down until it does.

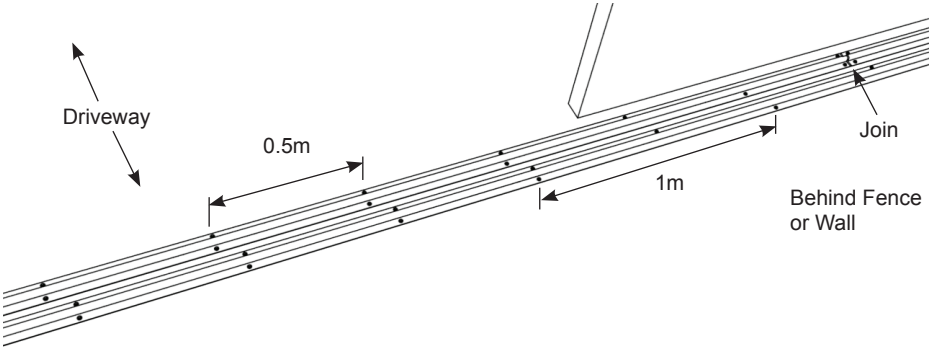
Check the gate is plumb in the open and close position and that the gate aligns with the keeper properly. You may need to adjust the position of the track to get this right depending on how plumb the post or wall is.



Adjust track if necessary to allow for post or wall not being plumb.

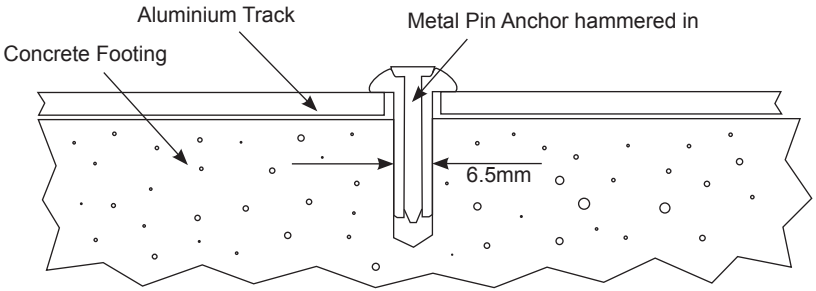
INSTALLING THE GATE

You can now fix the Guide Bracket and Keeper to the posts or walls in the exact position they were when clamped and you can fix the gate track to the concrete footing too. Mark the position the gate track is on the ground then mark out the position of the mounting holes on the track. You'll need to remove the gates to do this. Where the track can be driven on there should be a hole each side every half a metre or so. Where it's not driven on one hole each side every half a metre is fine. Where the track is joined there should be a hole each side on each end. Holes should be 8mm dia.



To drill the holes in the tracks lift them up on to blocks of wood, other wise the drill bit will blunten if it hits the concrete.

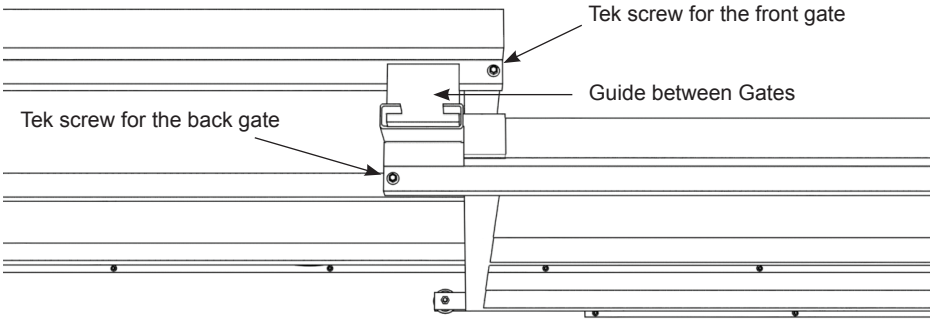
Then place the tracks back where initially marked and fix them to the concrete footing by drilling holes with a masonary drill bit into the concrete footing and hammer in the supplied metal pin anchors.



Metal pin anchors are much quicker to fit, hold better and look nicer than dyna bolts. You must use a 6.5mm dia. masonary drill bit to drill into the concrete then band them in with a hammer. A rotatary Hammer Drill is much faster for drilling concrete than a conventional hammer drill. Low cost rotary hammer dills are available that will save a lot of time if the concrete is hard.

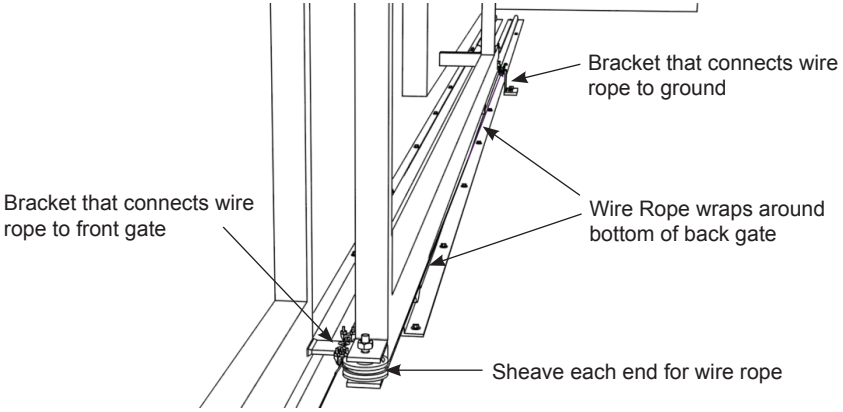
INSTALLING THE GATE

Place each gate back on its track as before then you are ready to fit timber. It's best to do this before going any further because you can still get access to all parts of the aluminium frame to screw timber palings on. Once this is done you should install tek screws onto the guide rails of each gate to stop them sliding out of their nylon guide blocks as the gates will now be a lot heavier and it will be more dangerous if they come out and fall over.



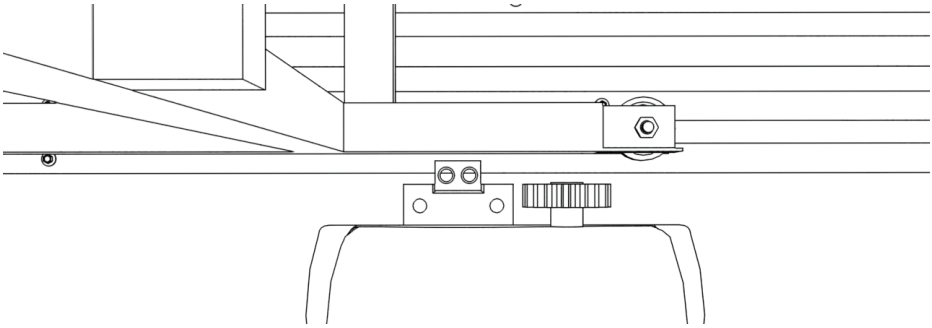
Now you're ready to fit the wire rope, which connects the gates together so moving the back gate will move the front gate at twice the speed.

The wire rope runs around the bottom of the back gate end to end connecting to the front gate at one point and the ground at another. Each end of the back gate has a sheave so the wire rope can move around it freely.

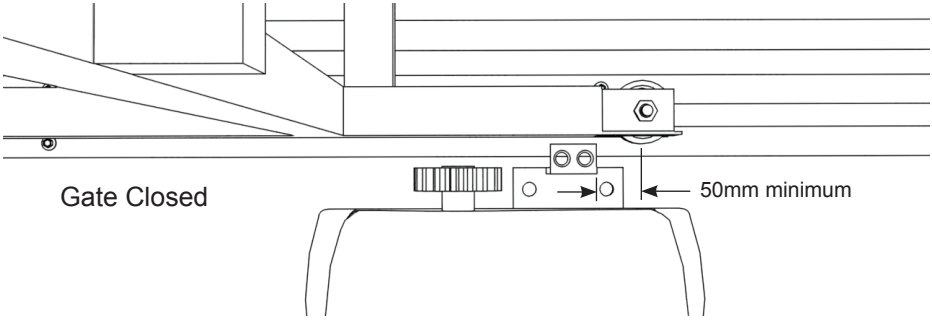


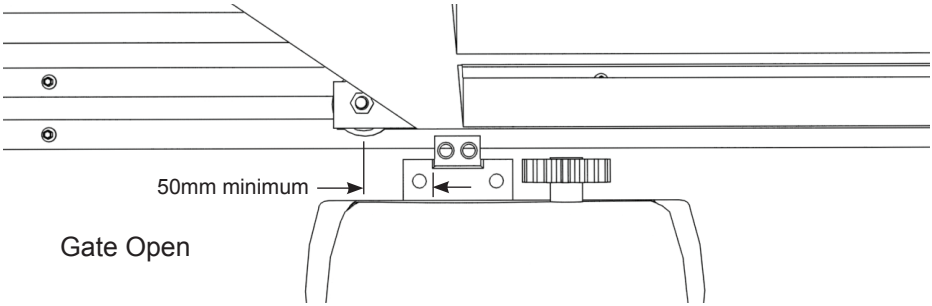
INSTALLING THE GATE

Placement of the ground connecting bracket for the wire rope is critical so it doesn't interfere with the automatic gate operator. If you place the automatic operator where it should go according to its installation instructions but don't fix it in place yet, you will be able to determine where the connecting bracket for the wire rope can go.



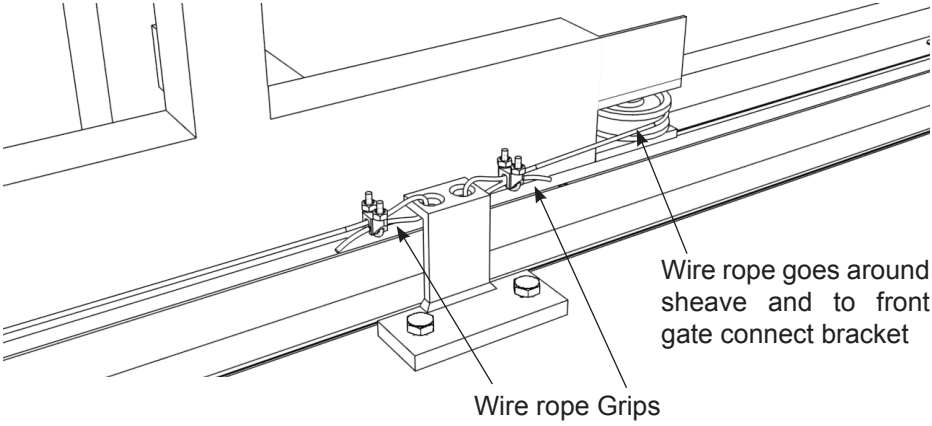
The connecting bracket must go as close as possible to the output gear for the gate operator and can go either side depending on where the output gear is positioned on the operator. There must be at least 50mm from either Sheave whether the gate is open or closed.



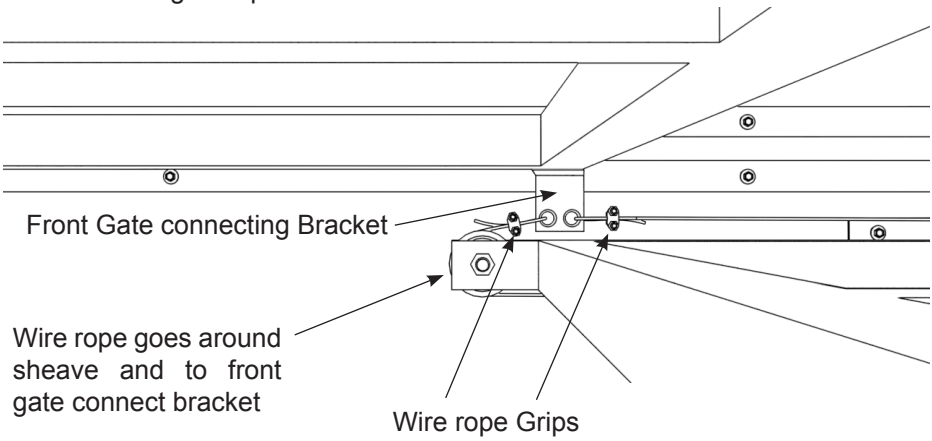


INSTALLING THE GATE

The connecting bracket should be dyna bolted to the concrete footing using the 10mm dyna bolts supplied. Then with the gate fully closed the wire rope can be fitted using the grips supplied. Wire rope can be cut with an angle grinder or hack saw.

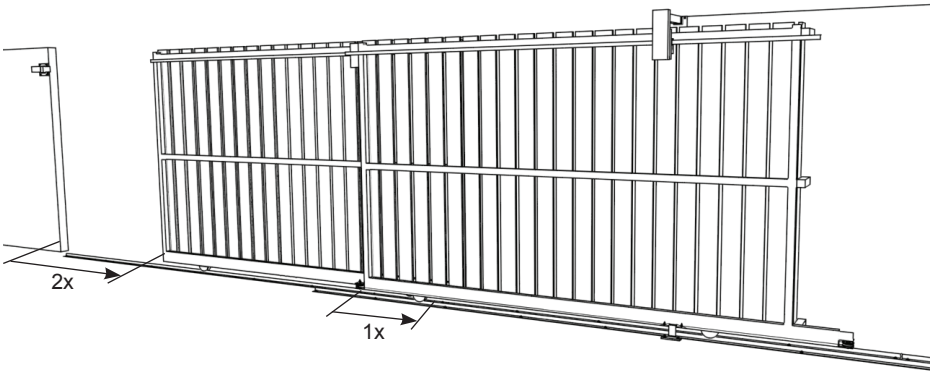


Install the wire rope to both connection brackets with the gate fully closed. Pull them as tight as possible.



INSTALLING THE GATE

Now when you open the back gate the front gate will be pulled open twice as much and the same will happen when closing.



You're now ready to install the automatic operator, which is done the same as a single piece sliding gate on the back gate only.